



FOREIGN AFFAIRS

Spy vs. AI

How Artificial Intelligence Will Remake Espionage

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In the early 1950s, the United States faced a critical intelligence challenge in its burgeoning competition with the Soviet Union. Outdated German reconnaissance photos from World War II could no longer provide sufficient intelligence about Soviet military capabilities, and existing U.S. surveillance capabilities were no longer able to penetrate the Soviet Union's closed airspace. This deficiency spurred an audacious moonshot initiative: the development of the U-2 reconnaissance aircraft. In only a few years, U-2 missions were delivering vital intelligence, capturing images of Soviet missile installations in Cuba and bringing near-real-time insights from behind the Iron Curtain to the Oval Office.

Today, the United States stands at a similar juncture. Competition between Washington and its rivals over the future of the global order is intensifying, and now, much as in the early 1950s, the United States must take advantage of its world-class private sector and ample capacity for innovation to outcompete its adversaries. The U.S. intelligence community must harness the country's sources of strength to deliver insights to

policymakers at the speed of today's world. The integration of artificial intelligence, particularly through large language models, offers groundbreaking opportunities to improve intelligence operations and analysis, enabling the delivery of faster and more relevant support to decisionmakers. This technological revolution comes with significant downsides, however, especially as adversaries exploit similar advancements to uncover and counter U.S. intelligence operations. With an AI race underway, the United States must challenge itself to be first—first to benefit from AI, first to protect itself from enemies who might use the technology for ill, and first to use AI in line with the laws and values of a democracy.

For the U.S. national security community, fulfilling the promise and managing the peril of AI will require deep technological and cultural changes and a willingness to change the way agencies work. The U.S. intelligence and military communities can harness the potential of AI while mitigating its inherent risks, ensuring that the United States maintains its competitive edge in a rapidly evolving global landscape. Even as it does so, the United States must transparently convey to the American public, and to populations and partners around the world, how the country intends to ethically and safely use AI, in compliance with its laws and values.

MORE, BETTER, FASTER

AI's potential to revolutionize the intelligence community lies in its ability to process and analyze vast amounts of data at unprecedented speeds. It can be challenging to analyze large amounts of collected data to generate time-sensitive warnings. U.S. intelligence services could leverage AI systems' pattern recognition capabilities to identify and alert human analysts to potential threats, such as missile launches or military movements, or important international developments that analysts know senior U.S. decisionmakers are interested in. This capability would ensure

that critical warnings are timely, actionable, and relevant, allowing for more effective responses to both rapidly emerging threats and emerging policy opportunities. Multimodal models, which integrate text, images, and audio, enhance this analysis. For instance, using AI to cross-reference satellite imagery with signals intelligence could provide a comprehensive view of military movements, enabling faster and more accurate threat assessments and potentially new means of delivering information to policymakers.

Intelligence analysts can also offload repetitive and time-consuming tasks to machines to focus on the most fulfilling work: generating original and deeper analysis, increasing the intelligence community's overall insights and productivity. A good example of this is foreign language translation. U.S. intelligence agencies invested early in AI-powered capabilities, and the bet has paid off. The capabilities of language models have grown increasingly sophisticated and accurate—OpenAI's recently released o1 and o3 models demonstrated significant progress in accuracy and reasoning ability—and can be used to even more quickly translate and summarize text, audio, and video files.

Although challenges remain, future systems trained on greater amounts of non-English data could be capable of discerning subtle differences between dialects and understanding the meaning and cultural context of slang or Internet memes. By relying on these tools, the intelligence community could focus on training a cadre of highly specialized linguists, who can be hard to find, often struggle to get through the clearance process, and take a long time to train. And of course, by making more foreign language materials available across the right agencies, U.S. intelligence services would be able to more quickly triage the mountain of foreign intelligence they receive to pick out the needles in the haystack that really matter.

The value of such speed to policymakers cannot be underestimated. Models can swiftly sift through intelligence data sets, open-source information, and traditional human intelligence and produce draft summaries or preliminary analytical reports that analysts can then validate and refine, ensuring the final products are both comprehensive and accurate. Analysts could team up with an advanced AI assistant to work through analytical problems, test ideas, and brainstorm in a collaborative fashion, improving each iteration of their analyses and delivering finished intelligence more quickly.

Consider Israel's experience in January 2018, when its intelligence service, the Mossad, covertly broke into a secret Iranian facility and stole about 20 percent of the archives that detailed Iran's nuclear activities between 1999 and 2003. According to Israeli officials, the Mossad collected some 55,000 pages of documents and a further 55,000 files stored on CDs, including photos and videos—nearly all in Farsi. Once the archive was obtained, senior officials placed immense pressure on intelligence professionals to produce detailed assessments of its content and whether it pointed to an ongoing effort to build an Iranian bomb. But it took these professionals several months—and hundreds of hours of labor—to translate each page, review it by hand for relevant content, and incorporate that information into assessments. With today's AI capabilities, the first two steps in that process could have been accomplished within days, perhaps even hours, allowing analysts to understand and contextualize the intelligence rapidly.

One of the most interesting applications is the way AI could transform how intelligence is consumed by policymakers, enabling them to interact directly with intelligence reports through ChatGPT-like platforms. Such capabilities would allow users to ask specific questions and receive summarized, relevant information from thousands of reports with source citations, helping them make informed decisions quickly.

Although AI offers numerous benefits, it also poses significant new risks, especially as adversaries develop similar technologies. China's advancements in AI, particularly in computer vision and surveillance, threaten U.S. intelligence operations. Because the country is ruled by an authoritarian regime, it lacks privacy restrictions and civil liberty protections. That deficit enables large-scale data collection practices that have yielded data sets of immense size. Government-sanctioned AI models are trained on vast amounts of personal and behavioral data that can then be used for various purposes, such as surveillance and social control. The presence of Chinese companies, such as Huawei, in telecommunications systems and software around the world could provide China with ready access to bulk data, notably bulk images that can be used to train facial recognition models, a particular concern in countries with large U.S. military bases. The U.S. national security community must consider how Chinese models built on such extensive data sets can give China a strategic advantage.

And it is not just China. The proliferation of "open source" AI models, such as Meta's Llama and those created by the French company Mistral AI and the Chinese company DeepSeek, is putting powerful AI capabilities into the hands of users across the globe at relatively affordable costs. Many of these users are benign, but some are not—including authoritarian regimes, cyber-hackers, and criminal gangs. These malign actors are using large language models to rapidly generate and spread false and malicious content or to conduct cyberattacks. As witnessed with other intelligence-related technologies, such as signals intercept capabilities and unmanned drones, China, Iran, and Russia will have every incentive to share some of their AI breakthroughs with client states and subnational groups, such as Hezbollah, Hamas, and the Wagner paramilitary company, thereby increasing the threat to the United States and its allies.

The U.S. military and intelligence community's AI models will become attractive targets for adversaries. As they grow more powerful and central to U.S. national security decision-making, intelligence AIs will become critical national assets that must be defended against adversaries seeking to compromise or manipulate them. The intelligence community must invest in developing secure AI models and in establishing standards for "red teaming" and continuous assessment to safeguard against potential threats. These teams can use AI to simulate attacks, uncovering potential weaknesses and developing strategies to mitigate them. Proactive measures, including collaboration with allies on and investment in counter-AI technologies, will be essential.

THE NEW NORMAL

These challenges cannot be wished away. Waiting too long for AI technologies to fully mature carries its own risks; U.S. intelligence capacities will fall behind those of China, Russia, and other powers that are going full steam ahead in developing AI. To ensure that intelligence—whether time-sensitive warnings or longer-term strategic insight—continues to be an advantage for the United States and its allies, the country's intelligence community needs to adapt and innovate. The intelligence services must quickly master the use of AI technologies and make AI a foundational element in their work. This is the only sure way to ensure that future U.S. presidents receive the best possible intelligence support, stay ahead of their adversaries, and protect the United States' sensitive capabilities and operations. Implementing these changes will require a cultural shift within the intelligence community. Today, intelligence analysts primarily build products from raw intelligence and data, with some support from existing AI models for voice and imagery analysis. Moving forward, intelligence officials should explore including a hybrid approach, in line with existing laws, using AI models trained on unclassified commercially available data and refined with classified

information. This amalgam of technology and traditional intelligence gathering could result in an AI entity providing direction to imagery, signals, open source, and measurement systems on the basis of an integrated view of normal and anomalous activity, automated imagery analysis, and automated voice translation.

To accelerate the transition, intelligence leaders must champion the benefits of AI integration, emphasizing the improved capabilities and efficiency it offers. The cadre of newly appointed chief AI officers has been established in U.S. intelligence and defense to serve as leads within their agencies for promoting AI innovation and removing barriers to the technology's implementation. Pilot projects and early wins can build momentum and confidence in AI's capabilities, encouraging broader adoption. These officers can leverage the expertise of national labs and other partners to test and refine AI models, ensuring their effectiveness and security. To institutionalize change, leaders should create other organizational incentives, including promotions and training opportunities, to reward inventive approaches and those employees and units that demonstrate effective use of AI.

The White House has created the policy needed for the use of AI in national security agencies. President Joe Biden's 2023 executive order regarding safe, secure, and trustworthy AI outlined the guidance needed to ethically and safely use the technology, and National Security Memorandum 25, issued in October 2024, is the country's foundational strategy for harnessing the power and managing the risks of AI to advance national security. Now, Congress will need to do its part. Appropriations are needed for departments and agencies to create the infrastructure needed for innovation and experimentation, conduct and scale pilot activities and assessments, and continue to invest in evaluation capabilities to ensure that the United States is constructing reliable and high-performing AI technologies.

Intelligence and military communities are committed to keeping humans at the heart of AI-assisted decision-making and have created the frameworks and tools to do so. Agencies will need guidelines for how their analysts should use AI models to make sure that intelligence products meet the intelligence community's standards for reliability. The government will also need to maintain clear guidance for handling the data of U.S. citizens when it comes to the training and use of large language models. It will be important to balance the use of emerging technologies with protecting the privacy and civil liberties of citizens. This means augmenting oversight mechanisms, updating relevant frameworks to reflect the capabilities and risks of AI, and fostering a culture of AI development within the national security apparatus that harnesses the potential of the technology while safeguarding the rights and freedoms that are foundational to American society.

Unlike the 1950s, when U.S. intelligence raced to the forefront of overhead and satellite imagery by developing many of the key technologies itself, winning the AI race will require that community to reimagine how it partners with private industry. The private sector, which is the primary means through which the government can realize AI progress at scale, is investing billions of dollars in AI-related research, data centers, and computing power. Given those companies' advancements, intelligence agencies should prioritize leveraging commercially available AI models and refining them with classified data. This approach enables the intelligence community to rapidly expand its capabilities without having to start from scratch, allowing it to remain competitive with adversaries. A recent collaboration between NASA and IBM to create the world's largest geospatial foundation model—and the subsequent release of the model to the AI community as an open-source project—is an exemplary demonstration of how this type of public-private partnership can work in practice.

As the national security community integrates AI into its work, it must ensure the security and resilience of its models. Establishing standards to deploy generative AI securely is crucial for maintaining the integrity of AI-driven intelligence operations. This is a core focus of the National Security Agency's new AI Security Center and its collaboration with the Department of Commerce's AI Safety Institute.

As the United States faces growing rivalry to shape the future of the global order, it is urgent that its intelligence agencies and military capitalize on the country's innovation and leadership in AI, focusing particularly on large language models, to provide faster and more relevant information to policymakers. Only then will they gain the speed, breadth, and depth of insight needed to navigate a more complex, competitive, and content-rich world.

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